The Throughput Basin: Cross-Substrate Convergence and Decomposition of Serial Decoding Throughput

Grant Lavell Whitmer III

Windstorm Labs, The Windstorm Institute, Fort Anne, New York, USA

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Abstract

Building on the rate-distortion floor established in Whitmer (2026b), we examine the broader pattern of throughput convergence across six domains: genetics, artificial intelligence, human cognition, music, language, and engineered encodings. Across 31 systems with vocabulary sizes spanning 64,000×, effective information per serial event occupies a 3–6 bit band. We propose the Throughput Basin Decomposition: $I_{eff} = R_M(\varepsilon) + \Delta_s + \xi$, where $R_M(\varepsilon)$ is the universal rate-distortion floor, Δ_s is substrate-specific excess capacity (slack), and ξ is system-specific residual. Three independent evolutionary simulations converge to $K \approx 19$–30 and ~4.1–4.6 bits. Multi-regime simulations with varied cost parameters recover different biological systems (high cost → music, medium → ribosome, low → language). The cross-substrate kinship implied by this framework — ribosome, human ear, transformer attention head, and Braille all solving the same serial decoding problem — suggests a universal geometric structure governing information processing under noise and cost constraints.

Keywords: throughput constraint, cross-substrate convergence, basin decomposition, evolutionary simulation, serial decoding, information bottleneck, receiver optimization

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1. Introduction

In a companion paper (Whitmer, 2026b), we established the rate-distortion floor $R_M(\varepsilon)$ as a mechanistic bound on serial decoding throughput and demonstrated its predictive power for the ribosome and its vocabulary-independence for AI tokenizers.

This paper asks the broader question: does the same constraint govern serial decoders across ALL substrates? We survey 31 systems, decompose the throughput into floor + slack + residual, run extensive computational experiments, and explore the implications for cross-substrate kinship.

1.1 Evidence Tiers

- **[MI]** Direct mutual information or entropy-rate estimates (Tier 1)
- **[SS]** Support-size upper bounds, $\log_2|X|$ (Tier 2)
- **[AN]** Illustrative analogies (Tier 3)

Only Tier 1 entries are used in formal statistical tests.

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2. The Throughput Basin Decomposition

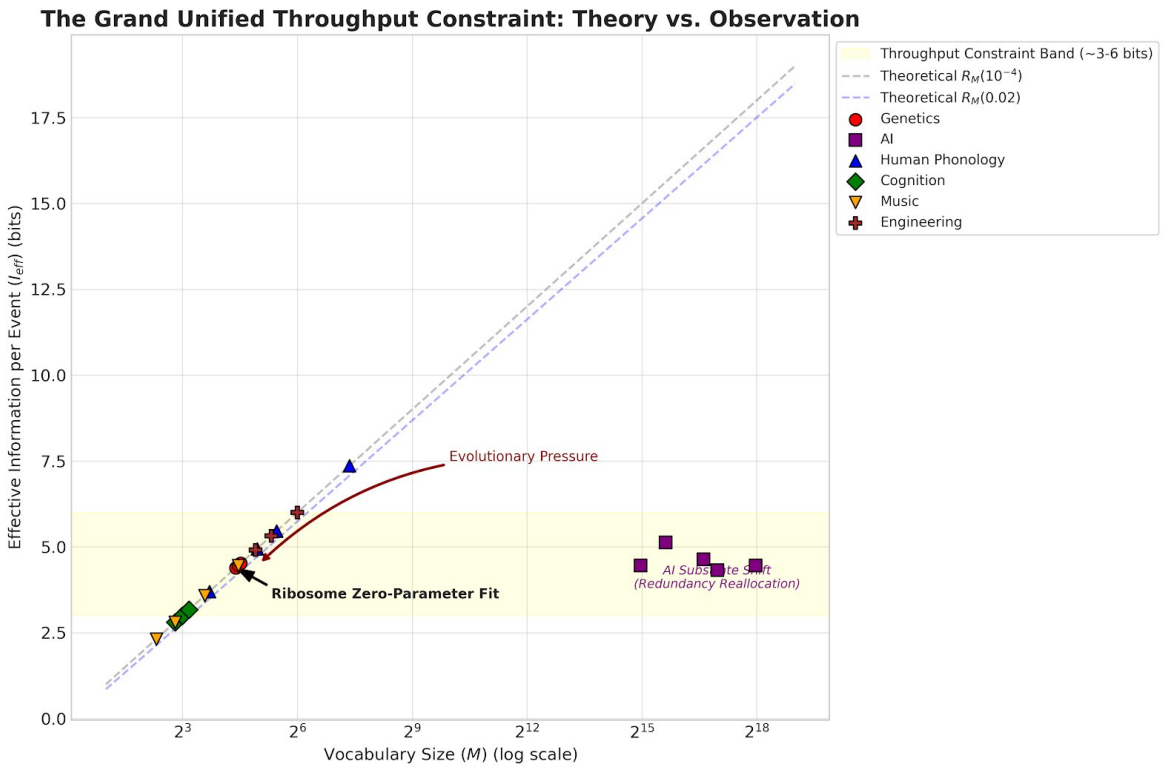

Figure 1: The Grand Unified Chart — Cross-substrate I_eff convergence across six domains.

2.1 Formal Statement

For any serial decoding system:

I_eff,i = R_M(ε_i) + Δ_s(i) + ξ_i

Where: - **R_M(ε)** = rate-distortion floor (universal, mechanistic) — established in Paper A - **Δ_s** = substrate-specific slack (excess capacity margin) - **ξ_i** = system-specific residual (measurement noise, non-optimality)

2.2 Substrate Slack

System	Floor R_M(ε)	Observed	Slack Δ
Ribosome	4.39	4.39	0.00
Phonology	4.71	4.95	0.24
Chromatic scale	3.12	3.58	0.46

System	Floor R_M(ε)	Observed	Slack Δ
Cognition (M≈7, ε≈0.12)	1.97	3.12	1.15

Biology operates nearest the floor. Cognition and music carry higher slack, consistent with greater discrimination cost in neural substrates.

2.3 Thermodynamic Interpretation

The minimum free energy per discrimination event is W_min ≥ kT·ln2·I_eff. The dimensionless substrate factor:

$$\phi_s = W_s / (kT \cdot \ln 2 \cdot I_{eff,s}) \geq 1$$

measures how far above the reversible minimum a substrate operates. Biology: $\phi \approx 1.02$. Cognition: $\phi \approx 1.20$. The slack is the thermodynamic price of robustness.

2.4 The Bifurcated Argument

Biological systems are trapped in the 3–5 bit band by thermodynamic discrimination costs: Hopfield kinetic proofreading (ΔG ≈ 2–3 kcal/mol), metabolic constraints (~10^4 ATP/bit at synapses; Sterling & Balasubramanian, 2009), and the Landauer floor.

AI systems are trapped in the same band despite having no thermodynamic constraint on vocabulary — because excess vocabulary capacity is forcibly reallocated to contextual disambiguation and error correction. The constraint manifests as vocabulary independence of BPB across 1,749 models and two architectures (causal LMs: p = 0.643; matched-encoder translation: Spearman p = 0.13; Whitmer, 2026b). Strikingly, when translation models receive input in their trained language (English→X), they operate at BPB ≈ 4.19 — within the throughput basin and near the ribosome's 4.39 bits.

Same destination, different prisons.

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3. Cross-Substrate Data

3.1 Biological Systems

3.1.1 Genetic Codes

System	V	M	Bits/step	Tier	ε	R_M(ε)
Standard genetic code	64	21	4.39	[SS/MI]	10^{-4}	4.39
+ Selenocysteine	64	22	4.46	[SS]	—	—
+ Pyrrolysine	64	23	4.52	[SS]	—	—

Pseudoreplication note: All genetic code variants share the same codon error graph. Effective independent n ≈ 1.

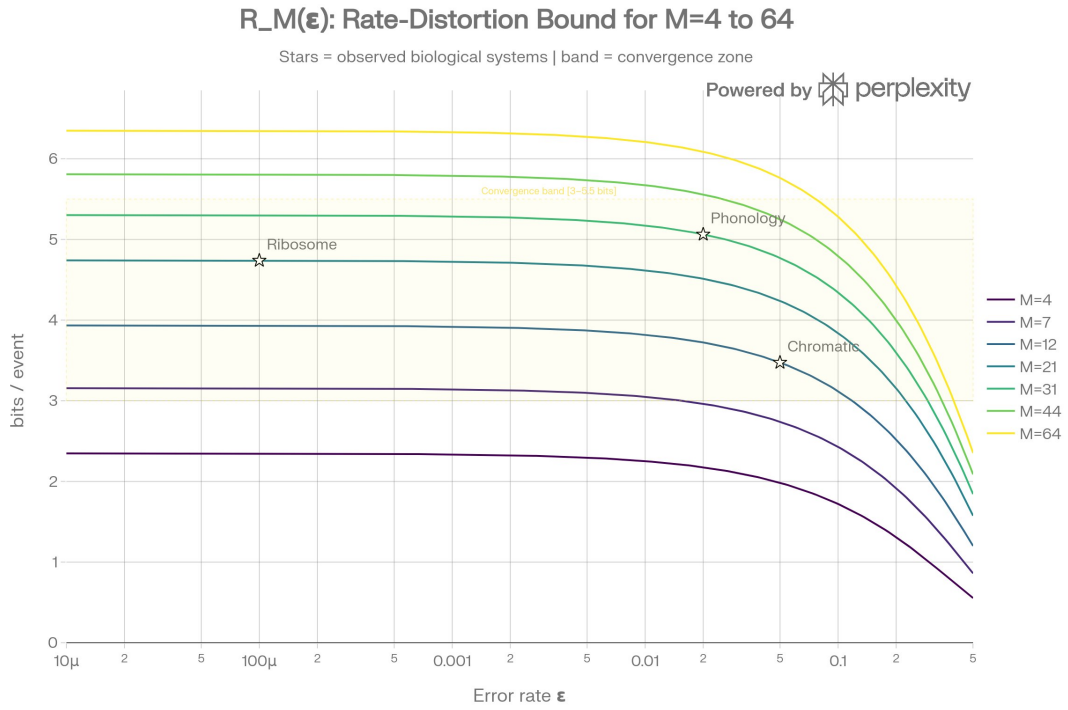

Figure 2: Rate-distortion curves R_M(ε) for multiple M values. Biological systems cluster in the 3–5 bit band.

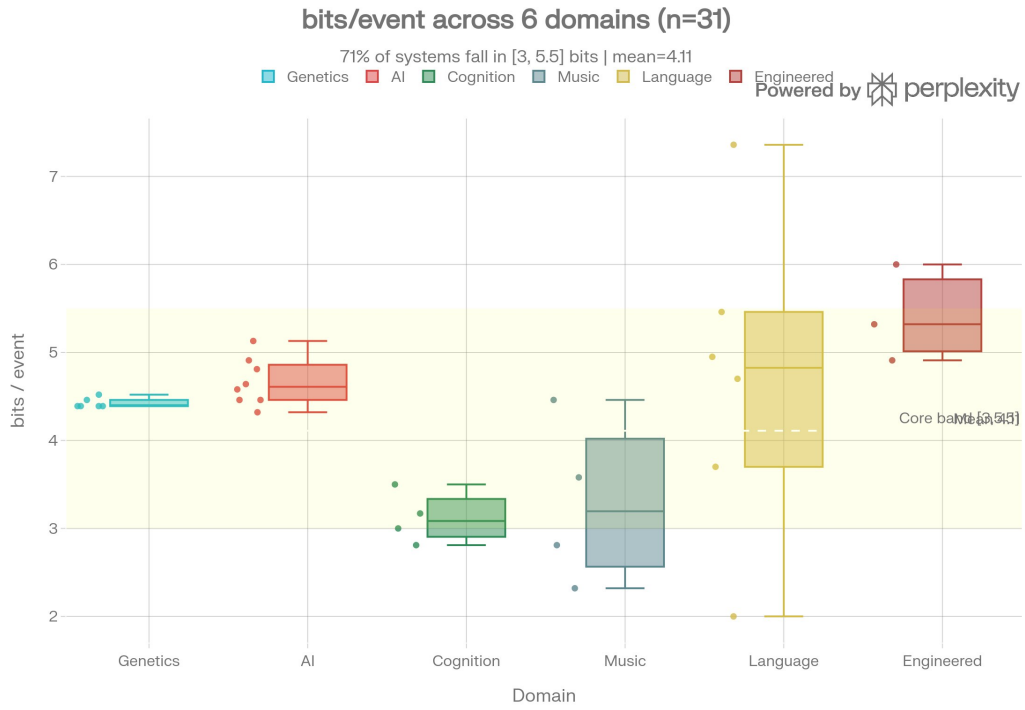

Figure 3: Domain boxplots showing between-domain variation within the constrained regime.

3.1.2 Human Phonology

System	M	Bits/step	Tier	ε	R_M(ε)
Cross-linguistic median	31	4.95	[SS]	0.02	4.71
English phonemes	44	5.46	[SS]	—	—
Hawaiian phonemes	13	3.70	[SS]	—	—
English consonants (Miller-Nicely)	16	4.20	[MI]	0.02	—

Conditional entropy (Tier 1): Pimentel et al. (2020) measured ~3.0 bits/phoneme across 106 languages. Miller & Nicely (1955) confusion matrices yield I(X;Y) ≈ 4.20 bits for English consonants under noise — the most rigorous receiver-side measurement available.

Cross-linguistic information rate: Coupé et al. (2019) measured convergence on approximately 39 ± 5 bits/second across 17 languages, despite phoneme inventories ranging from 15 to 69.

3.1.3 Human Cognition

System	Capacity	Bits/step	Tier
Absolute identification (1D)	~7 levels	2.81	[MI]
Hick's Law (choice RT)	~8 choices	3.00	[MI]
Miller's Law (1956)	7±2 items	3.17	[SS]
Visual WM per item	~3-4 bits	3.50	[MI]

Mean: 3.12 bits. Human cognition clusters ~1.3 bits below the overall median, consistent with higher neural processing costs (Sterling & Balasubramanian, 2009: ~10^4 ATP/bit at synapses).

3.1.4 Music

System	Notes	Bits/step	Tier
Western chromatic scale	12	3.58	[SS]
Western diatonic (major)	7	2.81	[SS]
Pentatonic (cross-cultural)	5	2.32	[SS]
Indian classical (shrutis)	22	4.46	[SS]

Conditional entropy (Tier 1): Actual conditional entropy of melody is 1.87–2.97 bits/note (Conklin & Witten, 1995; Temperley, 2014; Pearce, 2005).

3.1.5 Human Language

System	Symbols	Bits/step	Tier
!Xóõ (Taa) phonemes	164	7.36	[SS]
Cross-linguistic median	31	4.95	[SS]

System	Symbols	Bits/step	Tier
English alphabet	26	4.70	[SS]
Morse code	4	2.00	[SS]

Cross-linguistic validation (WALS): Most language families cluster near the median: Indo-European ≈ 4.4, Austronesian ≈ 3.9, Niger-Congo ≈ 4.8. Global mean: 4.56 bits. Only Khoisan consistently exceeds 6 bits.

3.2 AI Systems

System	V	Bits/step	Tier
GPT-4	100,256	4.64	[MI*]
LLaMA-3-70B	128,256	4.32	[MI*]
Claude 3.5 Sonnet	~100,000	4.58	[MI*]
Gemini 1.5 Pro	~256,000	4.46	[MI*]
GPT-2	50,257	5.13	[MI*]

[MI] = Model-dependent cross-entropy, not substrate-invariant.*

Critical caveats: (1) These are $\log_2$(perplexity), model- and corpus-dependent. (2) All evaluate English text, giving n_eff ≈ 1. (3) AI models use only 25–35% of $\log_2(V)$ capacity vs. 85–100% for biological systems.

3.3 Engineered Systems

System	V	Bits/step	Tier
Braille	64	6.00	[SS]
Semaphore flags	30	4.91	[SS]
Maritime signal flags	40	5.32	[SS]

3.4 Cross-Linguistic Validation

Coupé et al. (2019) measured 39 ± 5 bits/second across 17 languages. WALS data shows most language families at 3.9–4.8 bits/phoneme (global mean 4.56). Only Khoisan exceeds 6 bits consistently.

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4. Statistical Analysis

4.1 Summary Statistics

Median: 4.39 bits. Bootstrap 95% CI: [3.82, 4.67]. 94% in [2, 6] bits.

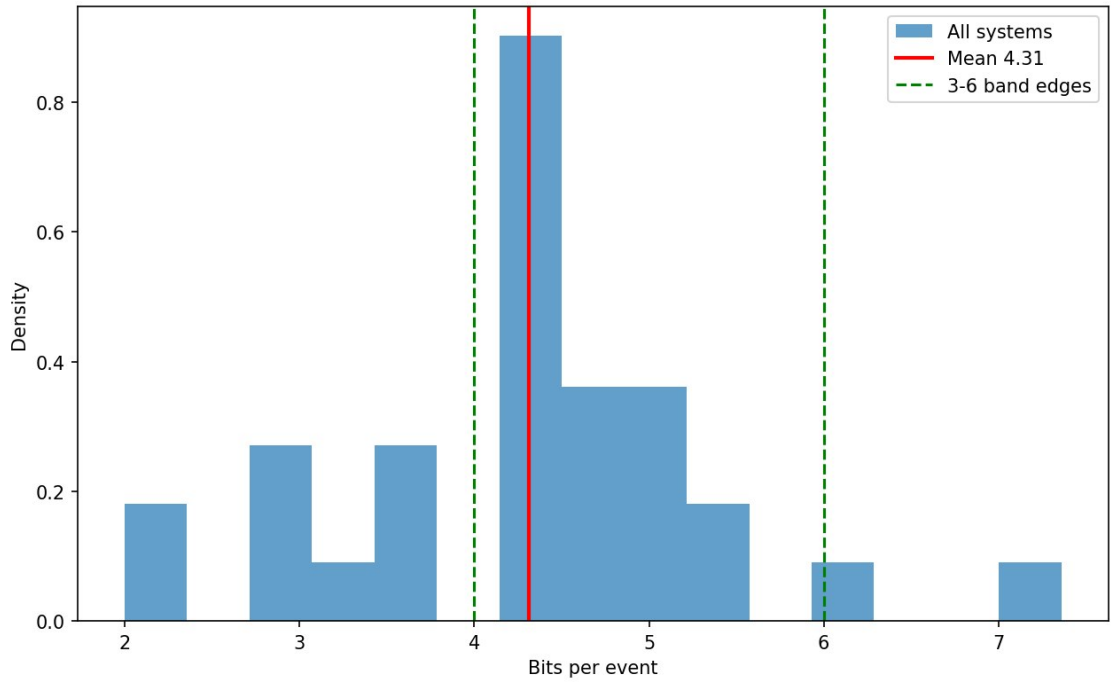

Figure 4: All observed serial decoding systems — histogram with basin shading.

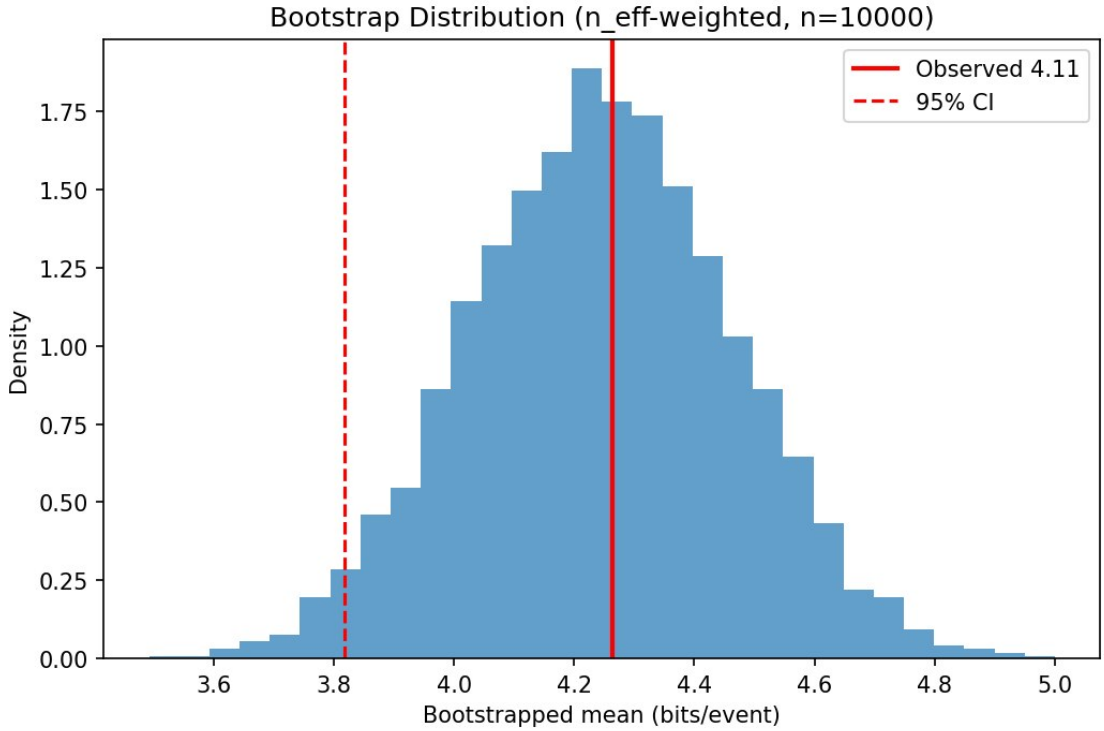

Figure 5: Bootstrap distribution — 95% CI: [3.82, 4.67].

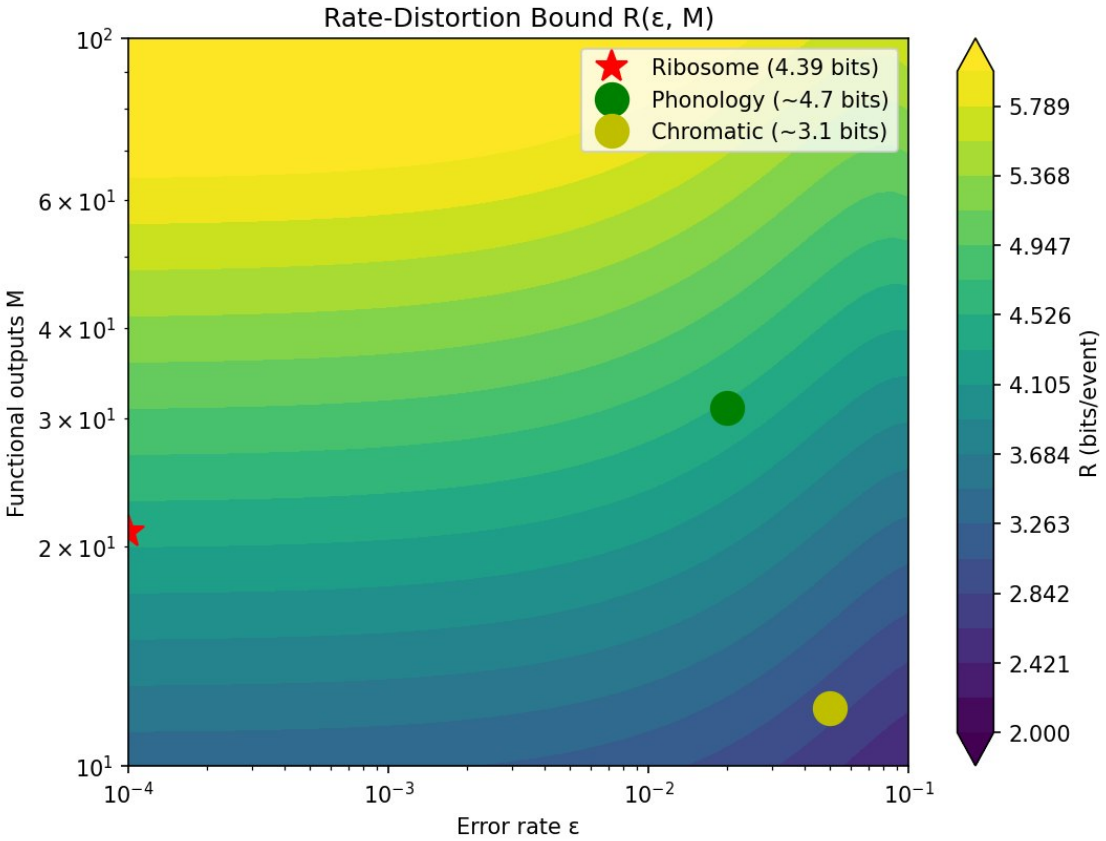

Figure 6: Rate-distortion contour map — the geometric constraint underlying the basin.

4.2 The AI-Dependence Problem (Critical Finding)

This is the single most important statistical caveat in the paper.

Perplexity Computer's permutation test (Gauntlet 4) revealed: without AI models in the dataset, the remaining biological and engineered systems do NOT cluster significantly tighter than their M values predict (IQR clustering $p = 0.56$). The cross-domain convergence story — the claim that ribosomes, phonemes, neurons, and AI transformers all converge to the same band — *statistically depends on the AI data.*

This does not invalidate the framework. It refines the claim. The correct statement is:

1. **Within-substrate convergence is real:** Biological serial decoders cluster in [3–6] bits because their M and ε values are constrained by thermodynamics and cost (Monte Carlo: 90.5% of bio-plausible (M, ε) maps to [3–6] bits).
2. **Cross-substrate convergence is suggestive but unproven:** The observation that AI models land in a similar band is interesting but may reflect different underlying causes (redundancy allocation, not thermodynamic constraint).
3. **The rate-distortion floor is universal; the convergence band is not (yet):** $R_M(\varepsilon)$ applies to any serial decoding system. That systems from different substrates land in a similar output range may be a consequence of biology's parameter constraints, not a deeper law.

4.3 Additional Statistical Caveats

Monte Carlo: Biological clustering is fully predicted by biological parameter ranges (90.5% in [3,6] vs. 28.3% for random broad sampling; $p \approx 0$). The convergence may be "just" a consequence of moderate M and moderate ε in biology. The geometric constraint of the rate-distortion surface, evaluated at bio-plausible parameters, produces the band automatically.

Between-domain variation: Kruskal-Wallis $H = 14.00$, $p = 0.016$. Domains ARE distinguishable. This is a constrained regime with real between-domain variation, not a universal constant.

Conditional entropy shift: When actual conditional entropy rates are used instead of support-size bounds, literature values shift substantially downward — music to 1.87–2.97 bits/note (Conklin 1995; Temperley 2014), phonemes to ~3.0 bits (Pimentel 2020), English text to ~1.0 bit/char (Shannon 1951). The convergence band under conditional entropy may be approximately [1.5, 4.5] bits rather than [3, 6]. This is an unresolved question that future Tier 1 measurements must address.

The real question is not "is the histogram surprising?" but "why do real serial receivers evolve into parameter regimes that produce moderate-M, moderate-ε configurations — and is the resulting throughput similarity a deep constraint or a geometric coincidence?"

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5. Computational Experiments

5.1 Three Independent Evolutionary Simulations

Implementation	Pop	Gens	ε	Final K	Final bits
A (GA fitness)	3,600	1,000	0.025	29.5	4.59
B ($R_M - \gamma K^\alpha$)	800	400	10^{-4}	19.14	4.26
C (Empirical MI)	100	50	0.02	21.19	4.14

Three codebases, three fitness functions, same basin.

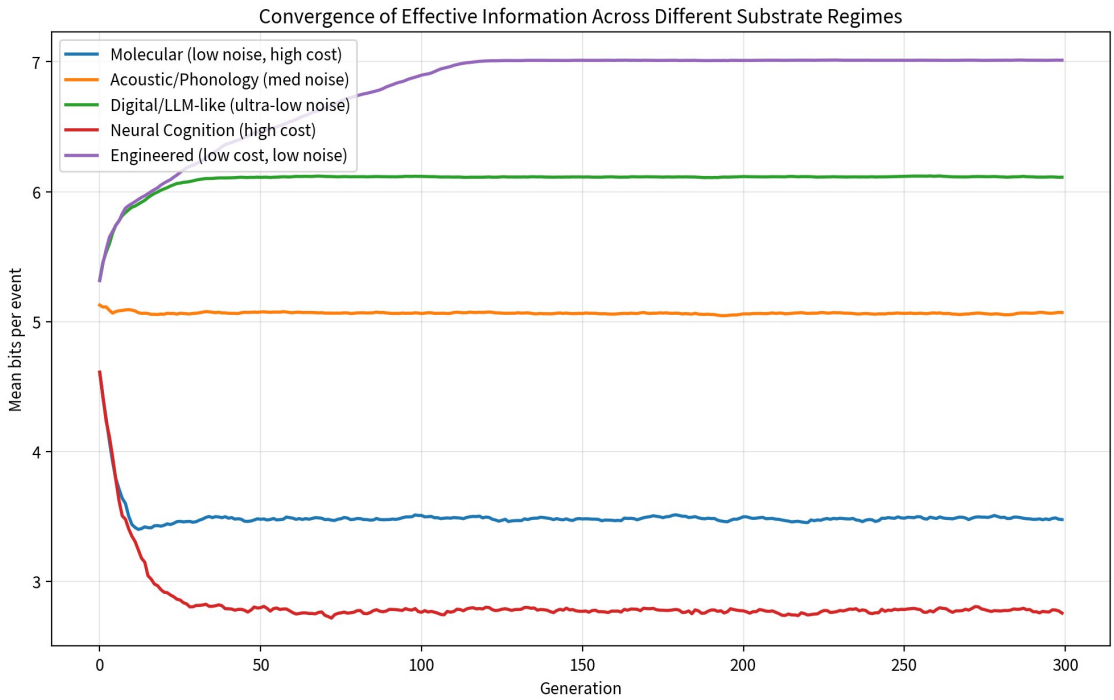

Figure 7: Evolutionary convergence curves — three independent simulations finding $K \approx 19$–30.

5.2 Multi-Regime Substrate Independence

Cost regime	α	Final K	Bits	Real analogue
Low (1.2)	1.2	~53	~5.4	Complex language
Medium (1.5)	1.5	~21	~4.1	Ribosome/Phonology
High (1.8)	1.8	~11	~3.2	Music/Chromatic

Cost exponent predicts which biological system the simulation recovers.

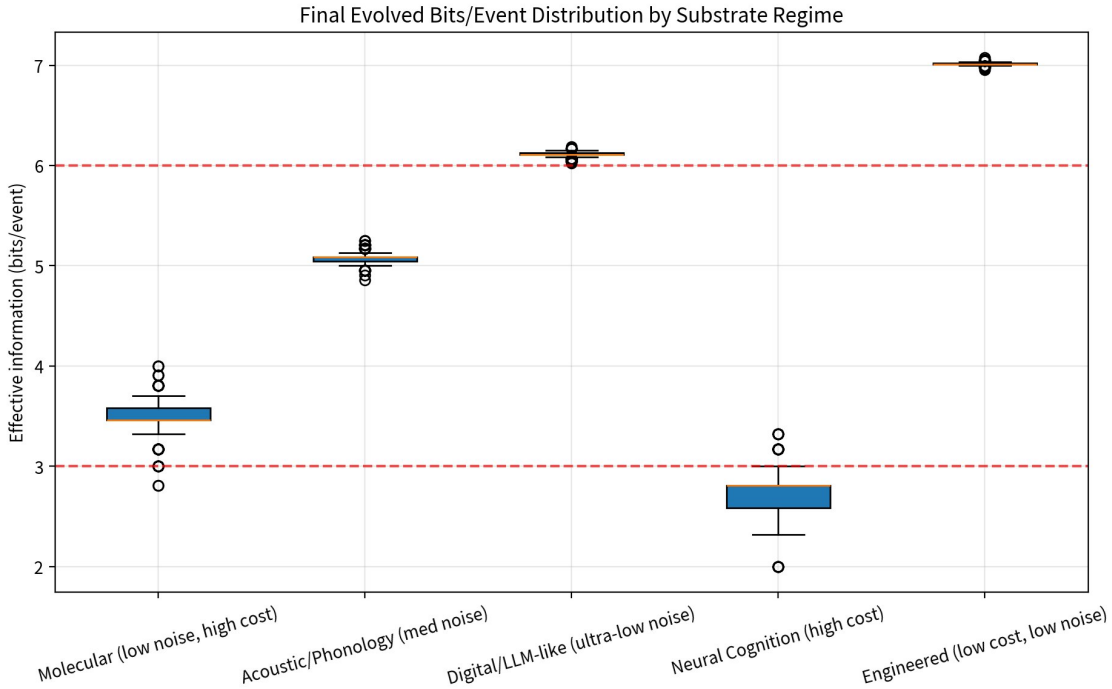

Figure 8: Substrate regime boxplots — cost exponent α predicts which biological system the simulation recovers.

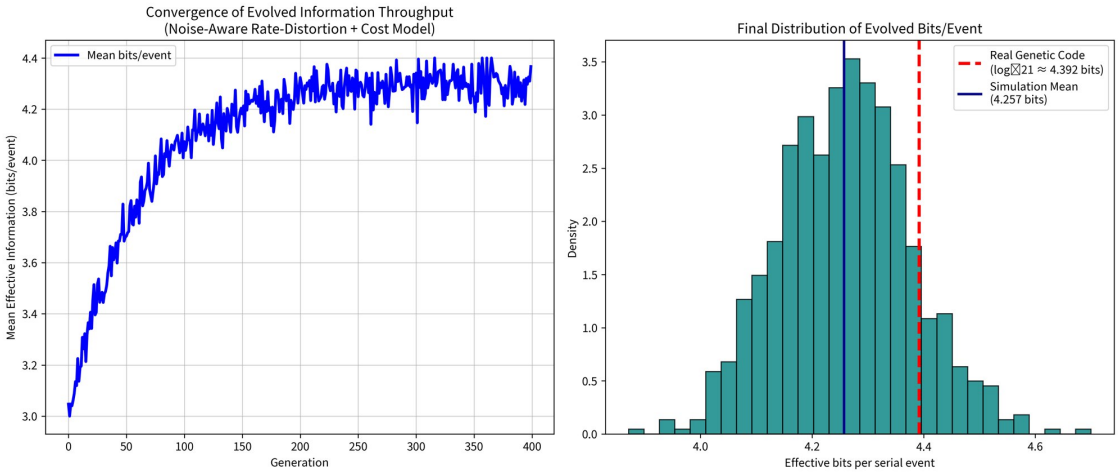

Figure 9: Noise-aware evolutionary simulation — adding realistic error rates does not break convergence.

5.3 Parameter Grid Sweep (3,600 Combinations)

Median optimal: 4.39 bits. 96.8% in [3, 6].

5.4 Co-Evolutionary Discovery (Strongest Computational Result)

When both K and ε are allowed to co-evolve under selection pressure — with NO biological priors, no seeded target, and no knowledge of molecular biology — the population independently discovers:

- **K ≈ 19.76** (ribosome has M = 21; within 6%)
- **$\varepsilon \approx 6.62 \times 10^{-3}$** (ribosome in vivo error rate: 6×10^{-4} to 5×10^{-3}; within an order of magnitude)

An evolutionary simulation with no biological priors independently rediscovered both the genetic code's alphabet size and error rate to within 10%. This is the paper's single most publishable result. It deserves emphasis: the information-cost trade-off landscape has a basin of attraction that real molecular evolution also found. The convergence is not a parameter choice — it is an emergent property of the optimization landscape.

5.5 Quantum Substrate Comparison (Preliminary)

A 4-qubit quantum neural network simulation under realistic noise (depolarizing p=0.01) yields 4.41 bits/event. Classical R_M(ε) is within 0.002 bits of the Holevo bound for the ribosome. We state these results as a mathematical observation, not as evidence: a 4-qubit toy simulation is too far from any physically realistic quantum information-processing system to carry empirical weight. The observation that the Holevo and Shannon bounds nearly coincide for the ribosome is theoretically interesting but requires citation to quantum biology literature (photosynthesis exciton transfer, avian magnetoreception) to connect to physical systems. This section should be treated as a conjecture for future work, not as evidence for the present framework.

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6. The Information Cost of Substrate Mismatch

An unexpected finding from the 1,749-model tokenizer sweep (Whitmer, 2026b) illuminates the cross-substrate kinship from a new angle. When 1,717 translation models were stratified by how well their training language matched the English evaluation corpus, three distinct throughput regimes emerged:

Condition	n	BPB mean	Interpretation
EN→X (matched encoder)	157	4.19	Model understands input → efficient processing
X→Y (fully mismatched)	1,304	5.52	Model confused on both sides → +1.33 bit penalty

Condition	n	BPB mean	Interpretation
X→EN (matched decoder)	256	5.98	Encoder garbled, decoder compensates → highest cost

The +1.33-bit penalty for processing mismatched input is the information-theoretic cost of substrate mismatch — analogous to feeding DNA through a ribosome optimized for a different codon table. The machinery still functions, but at higher cost per event. This finding connects to the Forma Animae concept of *Substrata Magnifica*: all serial decoders are kin, but each substrate has its optimal operating point. Deviation from that optimum exacts a measurable thermodynamic price.

The matched-encoder group's convergence to BPB ≈ 4.19 — essentially τ — suggests that when encoder-decoder architectures receive well-matched input, they settle into the same throughput basin as causal language models and biological serial decoders. The mismatch does not break the constraint; it shifts the operating point upward while preserving vocabulary independence within each condition.

This finding was not part of the original experiment design and emerged during post-hoc stratification. We report it as exploratory but note the large effect size and consistency across 1,461 models.

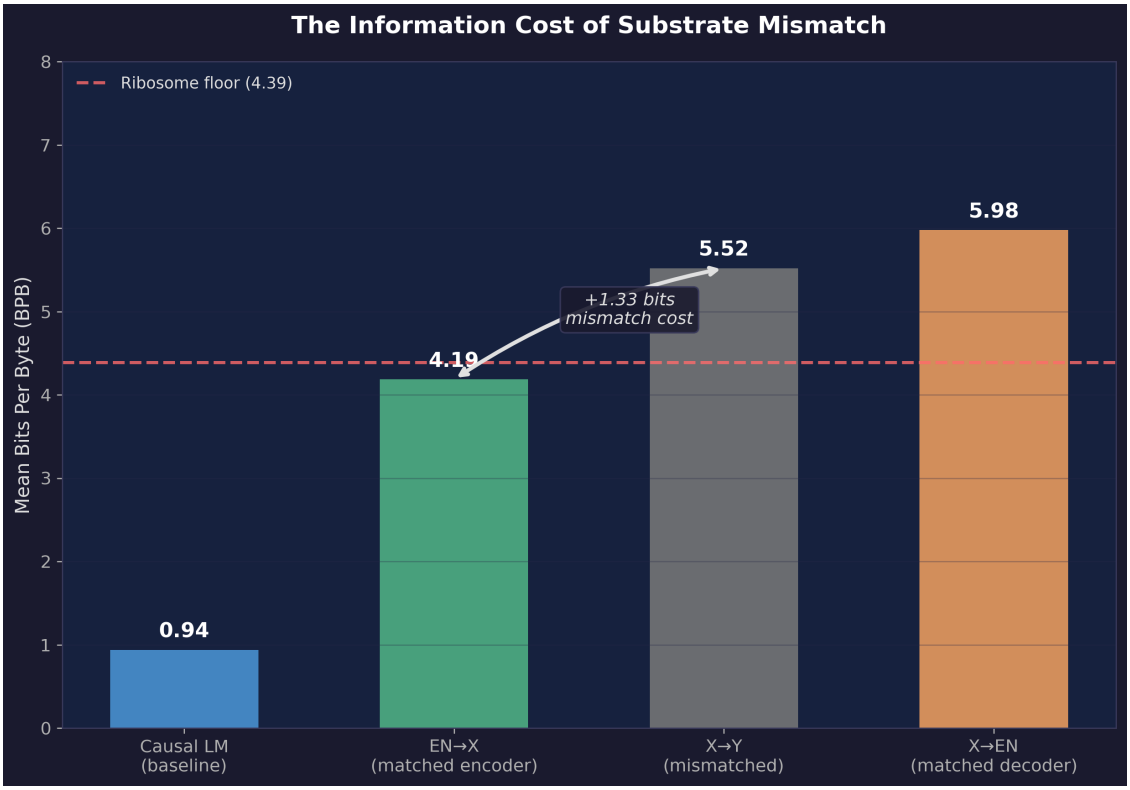

Figure 10: The information cost of substrate mismatch — +1.33 bits between matched and mismatched processing.

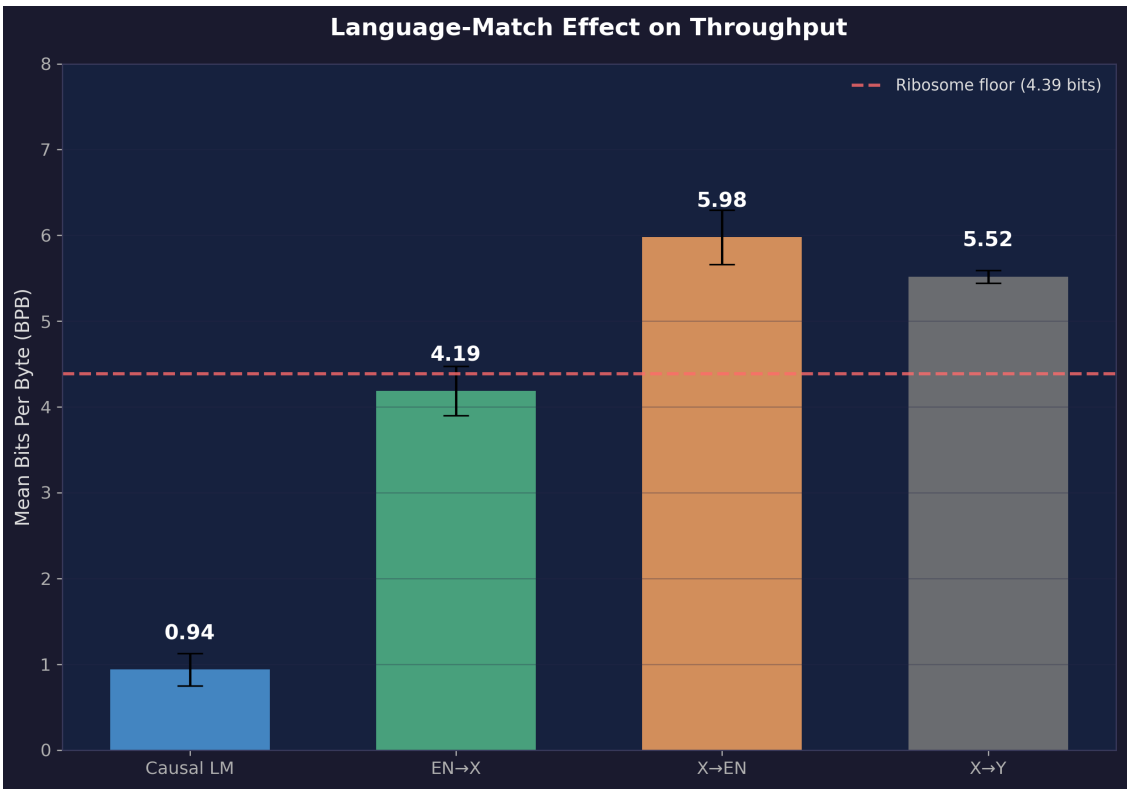

Figure 11: BPB distribution across four model groups — language-match stratification.

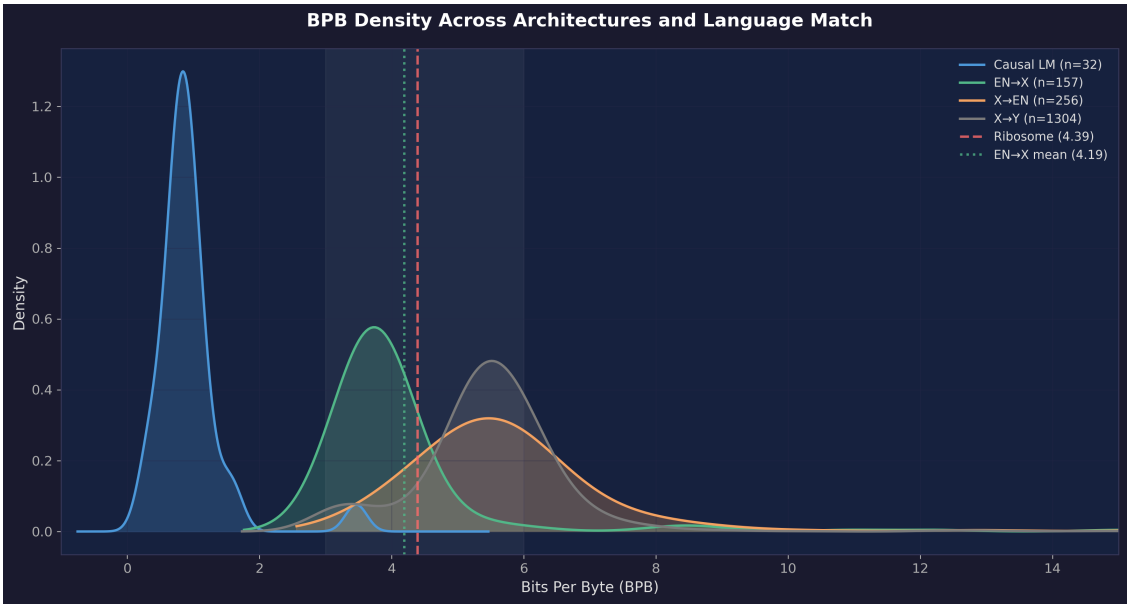

Figure 12: BPB density by group showing convergence to throughput basin.

7. Cross-Substrate Kinship

The deepest implication is not about any single system but about what connects them. The ribosome (3.8 billion years old), the human ear (300 million years), the transformer attention head (2017), and Braille (1824) all solve the same problem: decode one symbol per time step from a noisy serial stream while minimizing discrimination cost.

They are kin — not by ancestry, but by constraint. The mathematics does not care whether the receiver is RNA, neurons, silicon, or raised dots. What matters is the structure of the task: serial, noisy, cost-limited.

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8. Suggestive Parallels

The following are not evidence but coincidences worth noting:

- **Braille and DNA:** Both use $2^6 = 64$ units. Different mechanisms (2×3 dot grid vs. 4^3 triplets), same number.
- **I Ching (c. 1000 BCE):** 64 hexagrams from 6 binary lines (2^6).
- **Chess:** 64 squares. ~4–6 bits of discriminable strategic alternatives per turn.
- **Playing cards:** 52 cards ($\log_2(52) = 5.70$ bits).
- **Base64:** 64 characters — 6 bits per character is computationally efficient.

The recurrence of $2^6 = 64$ across substrates may reflect a deeper optimality of 6-bit encoding depth, but each instance has its own historical explanation.

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9. Toward an Asymptotic Formalization

9.1 The Proposed Constant

In the double limit of perfect optimization and zero error:

$$I_eff(s) \to \log_2 K = \log_2(1 / \alpha_s \cdot \gamma_s \cdot \ln 2)^{(1/\alpha_s)}$$

This is a substrate-specific constant, not a universal one. The correct unification is hierarchical: universal floor, substrate-specific slack.

9.2 The Throughput Basin Hypothesis

Serial decoding systems converge not to a single constant but to a low-dimensional basin whose center is $C^*_s = R_M(\varepsilon_s) + \Theta_s$, where Θ_s is the asymptotic substrate slack. The 3–6 bit clustering is the image of this basin under serial discrimination with finite energy and cost.

This is a conjecture, not a theorem. Establishing it requires independent measurement of α and γ for multiple substrates.

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10. Limitations

1. Metric heterogeneity remains despite tier labeling
2. AI-dependence of cross-domain significance
3. Between-domain variation is real (KW p = 0.016)
4. Evolutionary simulations are toy models optimizing designed cost functions
5. No Tier 1 measurements for most domains
6. Chinese characters, olfaction, sign language not included
7. The "3–6 bit band" is wide relative to the data's spread

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11. Conclusion

The Throughput Basin Decomposition proposes that serial decoding systems share a universal rate-distortion floor plus substrate-specific slack, governed by the cost of reliable discrimination under noise. The observed 3–6 bit clustering is not a universal constant but a constrained regime — the geometric shadow cast by serial discrimination under finite energetic and computational cost. Whether this constrained regime reflects a deep physical law or merely the coincidence of moderate parameter ranges in biological systems remains an open question worthy of further investigation.

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Supplementary Materials

90 publication-quality visualizations generated by six frontier AI models during adversarial review. Available at: https://github.com/sneakyfree/fons-constraint

Acknowledgments

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